

1) Mutant Generation

A Inputs

KernelBench

L1 + L2
200 tasks

LLM kernels

1,960 turns →
90 accepted

B Result Filtering

run correct

compile pass,
run wrong

compile wrong

keep top two classes;
discard last

C Operator Taxonomy

16 operators
in 4 categories

A · Arithmetic
B · GPU parallel
C · ML numerical
D · LLM patterns

D Mutant Synthesis

AST + regex
site matching

≤ 3 mutants /
(kernel × op)

1,646
mutants

2) Initial Testing

Initial Testing

torch.allclose vs
PyTorch reference
 $atol = rtol = 1e-2$

SURVIVED mutants

KILLED + STILLBORN

(closed; no signal)

3) Equivalent Mutant Detection (EMD)

Layer 0

CUDA + host source normalization
strip comments, collapse whitespace
otherwise

Layer 1

4 operator-aware static rules
*boundary_unreachable · dead_write ·
mask_no_reach · dead_host_constant*
otherwise

Layer 2

Dynamic bitwise testing (112 rounds)
*100 random + 12 directed; original vs
mutant CUDA*
otherwise

Layer 3

LLM review with Layer 0–2 evidence
*one-way safety: revoke only, cannot grant
equivalence*

STRICT_EQ
closed by proof

CANDIDATE_EQ
provisional

SURVIVED
bitwise divergence

CANDIDATE_EQ
L2 + L3 confirmed

SURVIVED
LLM revokes